

Hossam BOUDRAA



+33 6 48 04 44 53 | hossam.boudraa@outlook.com | [hossam-boudraa](https://www.linkedin.com/in/hossam-boudraa) | [hossamdb](https://github.com/hossamdb) | hossamdb.github.io

Multilingual NLP researcher-engineer specializing in large language models, document understanding, information extraction, and causal graph construction for scientific, biomedical, and social data.

Education

Master's Degree, Data Science and Artificial Intelligence

GALILÉE INSTITUTE – SORBONNE PARIS NORD UNIVERSITY

Paris, France

2021 – 2023

Master's Degree, Web Intelligence and Data Science (WISD)

SIDI MOHAMED BEN ABDELLAH UNIVERSITY

Fez, Morocco

2021 – 2023

Bachelor's Degree, Mathematics and Computer Science (SMI)

IBN ZOHR UNIVERSITY

Agadir, Morocco

2018 – 2021

Bachelor's Degree, French Studies

IBN ZOHR UNIVERSITY

Agadir, Morocco

2018 – 2021

Advanced Technician Diploma, Software Development

ISTA AGADIR

Agadir, Morocco

2016 – 2018

Technician Diploma, Computer Maintenance and Networking

ISTA AGADIR

Agadir, Morocco

2014 – 2016

Research Experience

Project Researcher

ATME — TAMPERE UNIVERSITY

Tampere, Finland

December 2025 – Present

- Fine-tune and evaluate large language models for domain-specific extraction and reasoning tasks.
- Build causal and equation-graph workflows to structure engineering knowledge.
- Contribute to research on neuro-symbolic integration of LLM outputs with structured knowledge representations.

Research Engineer

SESSTIM — AIX-MARSEILLE UNIVERSITY

Marseille, France

November 2023 – January 2026

- Processed data from the French National Health Data System (SNDS) for a public-health research project.
- Developed prediction algorithms and a recommendation system based on machine-learning techniques.
- Analyzed online forums to identify user profiles from unstructured text data.
- Co-authored peer-reviewed publications and contributed to grant-related deliverables.

Master's Research Internship

LIS LABORATORY — AIX-MARSEILLE UNIVERSITY

Marseille, France

March – September 2023

- Studied information transfer in French conversations under the supervision of Prof. Philippe Blache and Dr. Leonor Becerra-Bonache.
- Evaluated pretrained language models (FlauBERT, CamemBERT) for dialogue understanding.
- Developed an automatic conversation summarization system; results contributed to a LREC-COLING 2024 publication.

Software Developer

ÉCOLE NATIONALE D'ARCHITECTURE D'AGADIR

Agadir, Morocco

January – June 2021

- Design and development of a web application for academic grade management using PHP/Laravel.

Software Developer

PRÉFECTURE D'AGADIR

Agadir, Morocco

August 2018 – September 2019

- Development of a management application for rural associations' records in C#.

Software Developer

SOUSSNET

Agadir, Morocco

January – July 2018

- Development of a stock management application in C++.

Research Projects

EvalLLM2025 — Information Extraction Challenge

June 2025

- PARTICIPATED IN THE EVALLLM2025 CHALLENGE ON INFORMATION EXTRACTION IN THE HEALTH DOMAIN FROM FRENCH-LANGUAGE DOCUMENTS, IMPLEMENTING TWO MODEL-CENTRIC APPROACHES: A GENERATIVE GPT-4 APPROACH AND AN AUTOENCODER-BASED BERT MODEL.

NER Shared Task 2023 — Nested Named Entity Recognition in Arabic	July 2023
• DEVELOPED A NAMED ENTITY RECOGNITION SYSTEM FOR ARABIC USING MARBERT AND ARABERT ON THE WOJOOD DATASET.	
SemEval 2023 — Multilingual Tweet Hostility Analysis	January 2023
• DEVELOPED A MULTILINGUAL MODEL FOR HOSTILITY PREDICTION IN TWEETS, EVALUATED USING PEARSON’S R COEFFICIENT.	
iSarcasmEval — Sarcasm Detection in Arabic	December 2022
• FINE-TUNED MARBERT WITH FOCAL LOSS FOR INTENTIONAL SARCASM DETECTION IN ARABIC.	

Technical Skills

Programming Languages	Python, R, SQL, C++, JavaScript, Bash
DL & NLP Frameworks	PyTorch, TensorFlow, HuggingFace Transformers, spaCy, NLTK
Model Architectures	Transformers, BERT, GPT-4, RoBERTa, encoder-decoder, Seq2Seq, attention mechanisms
Methods	Representation learning, zero-shot & few-shot learning, prompt tuning, transfer learning, causal inference
Research Domains	Multilingual NLP, low-resource information extraction, hate speech detection, sarcasm detection, conversation analysis, biomedical NLP
Data Science	Pandas, NumPy, Seaborn, Matplotlib, Plotly, Jupyter, Google Colab, statistical analysis
Engineering	Flask, FastAPI, Streamlit, Django, REST APIs, Git, Linux scripting, Docker
Databases	PostgreSQL, MySQL, MongoDB, SQLite

Publications

WOAH @ ACL 2025 , <i>Implicit Hate Target Span Identification in Zero and Few-Shot Settings with Selective Sub-Billion Parameter Models</i>	July 2025
TALN 2025 , <i>Leveraging Prompting for Information Extraction in Low-Resource French Medical Texts</i>	July 2025
LREC-COLING 2024 , <i>Did You Get It? A Zero-Shot Approach to Locate Information Transfers in Conversations</i>	May 2024

Teaching

Post-Relational Databases — Teaching Fellow (part-time) MASTER’S IN COMPUTER SCIENCE, AIX-MARSEILLE UNIVERSITY	Marseille, France Sept 2025 – Dec 2025
Relational Database Systems — Teaching Fellow (part-time) POLYTECH MARSEILLE (3RD-YEAR ENGINEERING STUDENTS)	Marseille, France Sept 2025 – Dec 2025
Advanced Databases — Teaching Fellow (part-time) MASTER’S IN COMPUTER SCIENCE, AIX-MARSEILLE UNIVERSITY	Marseille, France Sept 2025 – Dec 2025
Post-Relational Databases — Teaching Fellow (part-time) MASTER’S IN COMPUTER SCIENCE, AIX-MARSEILLE UNIVERSITY	Marseille, France Sept 2024 – Dec 2024
Relational Database Systems — Teaching Fellow (part-time) POLYTECH MARSEILLE (3RD-YEAR ENGINEERING STUDENTS)	Marseille, France Sept 2024 – Dec 2024

Certifications

Deep Learning Specialization , Neural networks, CNNs and sequence models.	Coursera
Retrieval-Augmented Generation with Embeddings & Vectors , RAG pipelines, embeddings and vector databases.	Coursera
Data Engineering , Data pipelines, ETL, Airflow and BigQuery.	IBM
Advanced SQL , Advanced relational database manipulation.	HackerRank

Languages

French:	Fluent (reading, speaking, writing)
English:	Professional proficiency (reading, speaking, writing)
Spanish:	Conversational (reading, speaking)
German:	Basics
Finnish:	Basics